

SIMATS ENGINEERING

ASSESSMENT TOOL 1 – ENUMERATION ANALYSIS

Course Code: ITA14

Course Name: Ethical Hacking

Assessment Tool: 1 – Enumeration Analysis

CO3: Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities.

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QUESTION 1

Security Enumeration of scanme.nmap.org

Aim

To examine the publicly accessible host scanme.nmap.org using Nmap and determine its network availability, exposed TCP ports, running services, software versions, operating-system information, and additional information obtained through default NSE scripts.

Tools Required

- Kali Linux
- Nmap
- Terminal

Procedure

Step 1 – Verify Nmap Installation

```
nmap --version
```

Step 2 – Host Discovery

```
nmap -sn scanme.nmap.org
```

This checks whether the target is reachable.

Step 3 – Basic Port Scan

```
nmap scanme.nmap.org
```

Step 4 – Scan All TCP Ports

```
nmap -p- scanme.nmap.org
```

Step 5 – Identify Services and Versions

```
nmap -sV scanme.nmap.org
```

Step 6 – Operating System Detection

```
sudo nmap -O scanme.nmap.org
```

Step 7 – Default NSE Enumeration

```
nmap -sC scanme.nmap.org
```

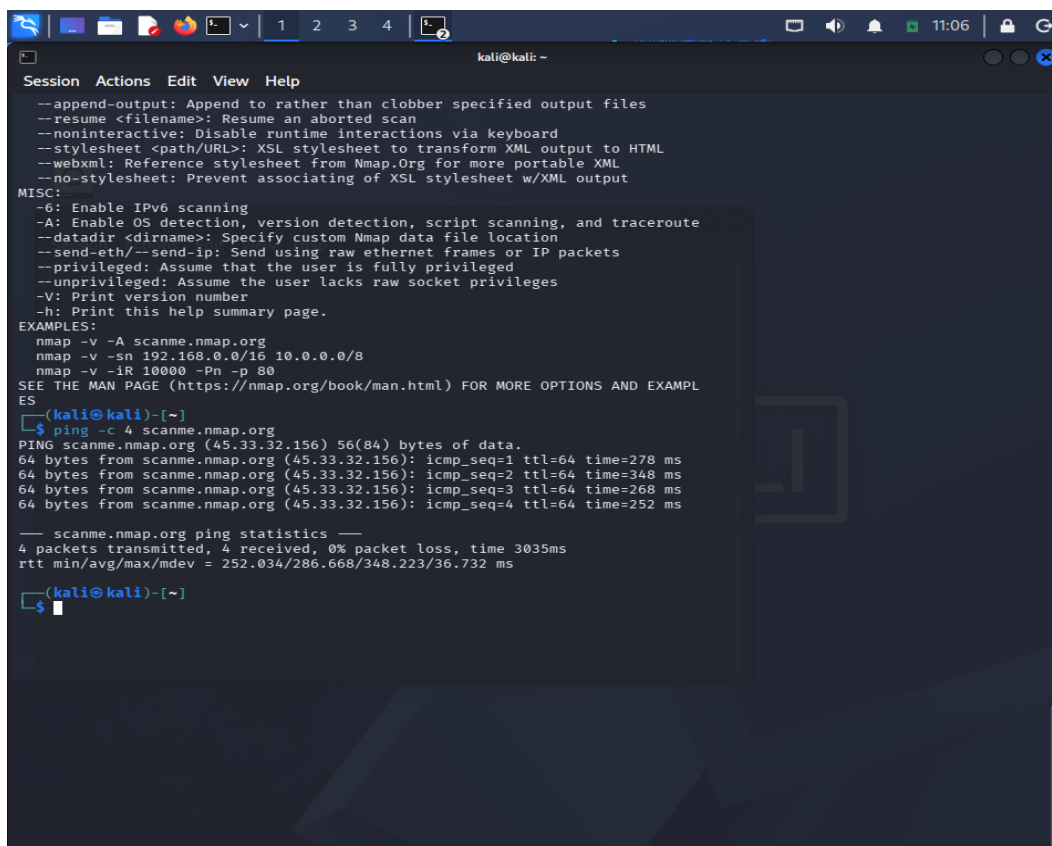
Step 8 – Combined Enumeration

```
sudo nmap -sC -sV -O scanme.nmap.org
```

Host Discovery Result

Target: scanme.nmap.org

Host Status:



```
kali@kali: ~  
Session Actions Edit View Help  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPL  
ES  
(kali@kali)-[~]  
$ ping -c 4 scanme.nmap.org  
PING scanme.nmap.org (45.33.32.156) 56(84) bytes of data.  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=1 ttl=64 time=278 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=2 ttl=64 time=348 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=3 ttl=64 time=268 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=4 ttl=64 time=252 ms  
— scanme.nmap.org ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3035ms  
rtt min/avg/max/mdev = 252.034/286.668/348.223/36.732 ms  
(kali@kali)-[~]  
$
```

Screenshot: Insert the host-discovery screenshot here.

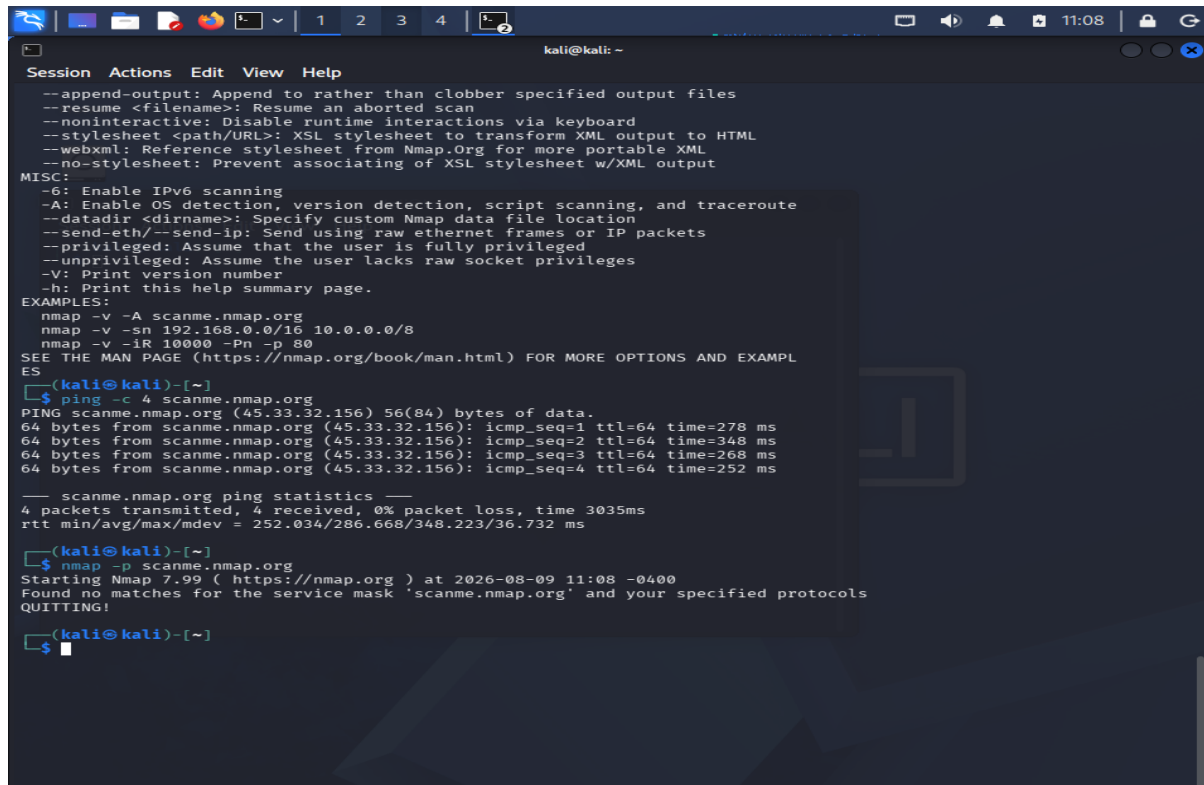
Port and Service Results

Port	State	Service	Version
[Actual]	Open	[Actual]	[Actual]
[Actual]	Open	[Actual]	[Actual]
[Actual]	Open	[Actual]	[Actual]

Interpretation

The port scan provides information about the services that can be reached from the network. Version detection gives additional information about the software operating behind each open port. Such information is useful during an authorized security assessment because it helps determine which services require patching, configuration review, or access restrictions.

Operating-system fingerprinting provides an estimate of the underlying platform. Default NSE scripts provide additional service-related information that may not be visible through a simple port scan.



```
kali@kali: ~  
Session Actions Edit View Help  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES  
(kali@kali)-[~]  
$ ping -c 4 scanme.nmap.org  
PING scanme.nmap.org (45.33.32.156): 56(84) bytes of data:  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=1 ttl=64 time=278 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=2 ttl=64 time=348 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=3 ttl=64 time=268 ms  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=4 ttl=64 time=252 ms  
— scanme.nmap.org ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3035ms  
rtt min/avg/max/mdev = 252.034/286.668/348.223/36.732 ms  
(kali@kali)-[~]  
$ nmap -p scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 11:08 -0400  
Found no matches for the service mask 'scanme.nmap.org' and your specified protocols  
QUITTING!  
(kali@kali)-[~]  
$
```

```
kali@kali: ~  
Session Actions Edit View Help  
--max-rate <number>: Send packets no faster than <number> per second  
FIREWALL/IDS EVASION AND SPOOFING:  
-f: --mtu <val>: fragment packets (optionally w/given MTU)  
-D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys  
-S <IP_Address>: Spoof source address  
-e <iface>: Use specified interface  
-g/--source-port <portnum>: Use given port number  
--proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies  
--data <hex string>: Append a custom payload to sent packets  
--data-string <string>: Append a custom ASCII string to sent packets  
--data-length <num>: Append random data to sent packets  
--ip-options <options>: Send packets with specified ip options  
--ttl <val>: Set IP time-to-live field  
--spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address  
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum  
OUTPUT:  
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, sI<br>pt kIddi3,  
and Grepable format, respectively, to the given filename.  
-oA <basename>: Output in the three major formats at once  
-v: Increase verbosity level (use -vv or more for greater effect)  
-d: Increase debugging level (use -dd or more for greater effect)  
--reason: Display the reason a port is in a particular state  
--open: Only show open (or possibly open) ports  
--packet-trace: Show all packets sent and received  
--iflist: Print host interfaces and routes (for debugging)  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -IR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPL  
ES  
$ ping -c 4 scanme.nmap.org  
PING scanme.nmap.org (45.33.32.156) 56(84) bytes of data:  
64 bytes from scanme.nmap.org (45.33.32.156): icmp_seq=1 ttl=64 time=278 ms
```

Attack Surface Assessment

The combination of open ports, service names, versions, and OS information forms the host's observable attack surface. An unauthorized party could use this information to research weaknesses associated with particular software versions or identify services that require authentication.

For example, an exposed network service could be investigated for configuration weaknesses, authentication problems, or publicly documented vulnerabilities. Therefore, only services required for legitimate public functionality should remain accessible.

Recommendations

- Remove unnecessary Internet-facing services.
- Keep all exposed software updated.
- Restrict administrative services to trusted networks.
- Configure firewall rules according to business requirements.
- Use strong authentication for administrative access.
- Monitor exposed services and authentication attempts.
- Conduct periodic authorized security reviews.

Conclusion

The Nmap assessment successfully demonstrates how network enumeration can be used to build an understanding of a publicly accessible host. The information obtained should be used to reduce unnecessary exposure and improve the security configuration of the server.

QUESTION 2

Enumeration of Metasploitable 2 – 192.168.56.101

Aim

To examine the deliberately vulnerable Metasploitable 2 Linux system in the isolated laboratory and identify its accessible network services, versions, operating-system information, and potential security weaknesses.

Procedure

Step 1 – Check Connectivity

```
ping -c 4 192.168.56.101
```

Step 2 – Host Discovery

```
nmap -sn 192.168.56.101
```

Step 3 – Standard Port Scan

```
nmap 192.168.56.101
```

Step 4 – Full TCP Port Scan

```
nmap -p- 192.168.56.101
```

Step 5 – Service Detection

```
nmap -sV 192.168.56.101
```

Step 6 – OS Fingerprinting

```
sudo nmap -O 192.168.56.101
```

Step 7 – NSE Default Scripts

```
nmap -sC 192.168.56.101
```

Step 8 – Comprehensive Enumeration

```
sudo nmap -sC -sV -O 192.168.56.101
```

Results

Target: 192.168.56.101

Host Status:

```
kali@kali: ~  
Session Actions Edit View Help  
--ip-options <options>: Send packets with specified ip options  
--ttl <val>: Set IP time-to-live field  
--spooof-mac <mac address/prefix/vendor name>: Spoof your MAC address  
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum  
OUTPUT:  
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, s|<riPt kIdDi3,  
and Grepable format, respectively, to the given filename.  
-oA <basename>: Output in the three major formats at once  
-v: Increase verbosity level (use -vv or more for greater effect)  
-d: Increase debugging level (use -dd or more for greater effect)  
--reason: Display the reason a port is in a particular state  
--open: Only show open (or possibly open) ports  
--packet-trace: Show all packets sent and received  
--iflist: Print host interfaces and routes (for debugging)  
--append-output: Append to rather than clobber specified output files  
--resume <filename>: Resume an aborted scan  
--noninteractive: Disable runtime interactions via keyboard  
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML  
--webxml: Reference stylesheet from Nmap.Org for more portable XML  
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output  
MISC:  
-6: Enable IPv6 scanning  
-A: Enable OS detection, version detection, script scanning, and traceroute  
--datadir <dirname>: Specify custom Nmap data file location  
--send-eth/--send-ip: Send using raw ethernet frames or IP packets  
--privileged: Assume that the user is fully privileged  
--unprivileged: Assume the user lacks raw socket privileges  
-V: Print version number  
-h: Print this help summary page.  
EXAMPLES:  
nmap -v -A scanme.nmap.org  
nmap -v -sn 192.168.0.0/16 10.0.0.0/8  
nmap -v -iR 10000 -Pn -p 80  
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPL  
ES  
[kali@kali]~  
$ nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcrc2-  
10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6  
Compiled without:  
Available nsock engines: epoll poll select  
[kali@kali]~  
$ ping -c 4.192.168.56.101  
ping: invalid argument: '4.192.168.56.101'  
[kali@kali]~  
$ ping -c 4 194.168.56.101
```

Service Inventory

Port	Service	Version	Function	Risk
[Actual]	[Actual]	[Actual]	[Function]	High/Medium/Low
[Actual]	[Actual]	[Actual]	[Function]	High/Medium/Low
[Actual]	[Actual]	[Actual]	[Function]	High/Medium/Low

Risk Evaluation

The discovered services should be categorized according to their purpose and exposure.

High Risk: Services that expose sensitive remote access, authentication, file-sharing, database, or other critical functionality, especially when poorly secured or outdated.

Medium Risk: Services that may be required but should have restricted access, strong authentication, and appropriate configuration.

Low Risk: Services that are necessary and properly secured with limited exposure.

```
kali@kali: ~  
Session Actions Edit View Help  
Nmap scan report for 192.168.56.101  
Host is up (0.00095s latency).  
All 1000 scanned ports on 192.168.56.101 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.82 seconds  
  
kali@kali: ~  
$ sudo nmap -sC -sV -O 192.168.56.101 -oN metasploitable_report.txt  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 12:00 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.00088s latency).  
All 1000 scanned ports on 192.168.56.101 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: 3Com 4500G switch (92%), AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), H3C Comware 5.20 (92%), HP Tru64 UNIX 5.1A (92%), Huawei VRP 8.100 (92%), Microsoft Windows Server 2003 SP1 (92%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (92%), Sanyo PLC-XU88 digital video projector (92%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 14.75 seconds  
  
kali@kali: ~  
$ sudo nmap -sC -sV -O 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-09 12:02 -0400  
Nmap scan report for 192.168.56.101  
Host is up (0.00093s latency).  
All 1000 scanned ports on 192.168.56.101 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 12.86 seconds  
  
kali@kali: ~  
$
```

Security Assessment

The enumeration results allow a penetration tester to determine which services should receive additional investigation. Instead of testing the entire system without prioritization, the tester can focus on the services that represent the greatest potential security concern.

The version information can also be compared with vendor security information during an authorized assessment.

Recommendations

- Disable services that have no legitimate purpose.
- Restrict remote-access services.
- Apply security updates.
- Use strong authentication.
- Implement host-based firewall controls.
- Restrict database and file-sharing services.
- Monitor suspicious connection attempts.
- Separate vulnerable laboratory systems from production networks.

Conclusion

The Metasploitable 2 exercise demonstrates the value of enumeration during the reconnaissance stage. Identifying the available services and their versions helps security professionals understand the system's exposure and prioritize subsequent authorized security testing.

QUESTION 3

Security Assessment of Windows Workstation – 192.168.56.102

Aim

To evaluate the network exposure of the Windows workstation before production deployment and determine whether the available services comply with the principle of least privilege.

Commands Used

```
ping -c 4 192.168.56.102
nmap -sn 192.168.56.102
nmap 192.168.56.102
nmap -p- 192.168.56.102
nmap -sV 192.168.56.102
sudo nmap -O 192.168.56.102
nmap -sC 192.168.56.102
sudo nmap -sC -sV -O 192.168.56.102
```

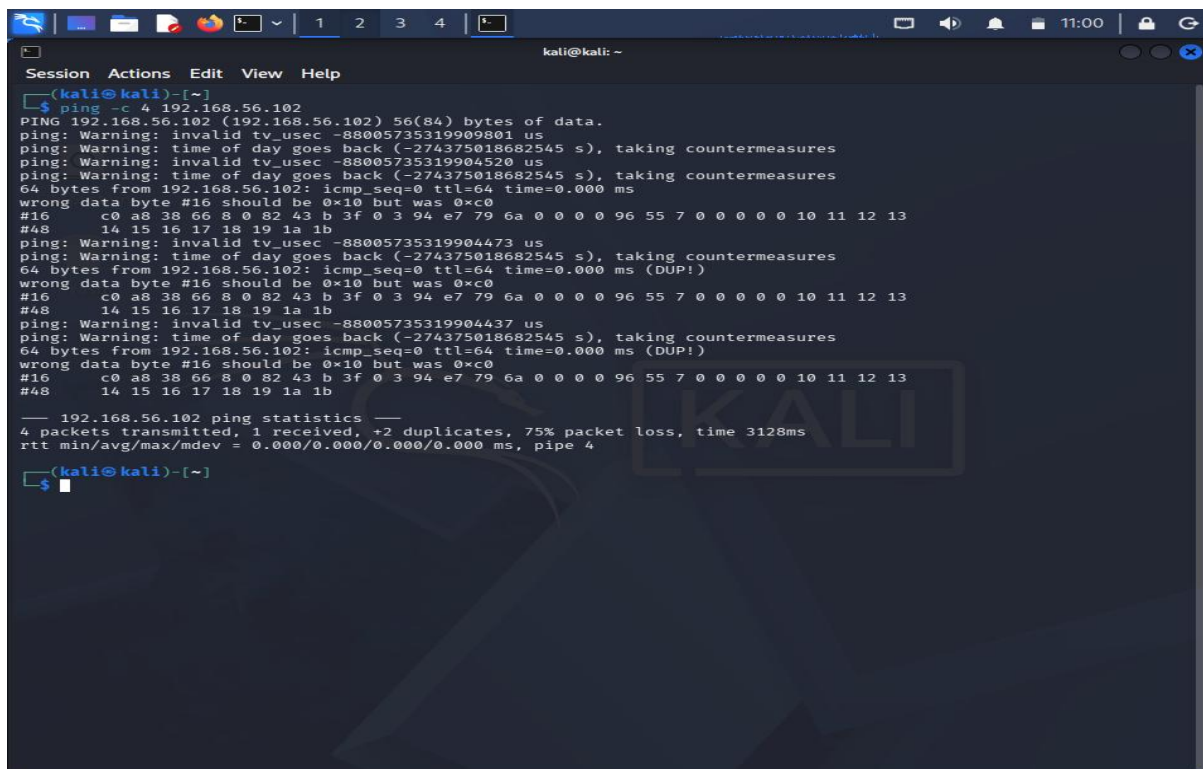
Initial Observation

The first scan was used to determine whether the workstation was reachable from the Kali system.

Host: 192.168.56.102

Enumeration Results

Port	Protocol	Service	Version	Purpose	Risk
[Actual]	TCP/UDP	[Actual]	[Actual]	[Purpose]	[Level]
[Actual]	TCP/UDP	[Actual]	[Actual]	[Purpose]	[Level]
[Actual]	TCP/UDP	[Actual]	[Actual]	[Purpose]	[Level]



```
kali@kali: ~  
Session Actions Edit View Help  
$ ping -c 4 192.168.56.102  
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.  
ping: Warning: invalid tv_usec -88005735319909801 us  
ping: Warning: time of day goes back (-274375018682545 s), taking countermeasures  
ping: Warning: invalid tv_usec -88005735319904520 us  
ping: Warning: time of day goes back (-274375018682545 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms  
wrong data byte #16 should be 0x10 but was 0xc0  
#16  c0 a8 38 66 8 0 82 43 b 3f 0 3 94 e7 79 6a 0 0 0 0 96 55 7 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319904473 us  
ping: Warning: time of day goes back (-274375018682545 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x10 but was 0xc0  
#16  c0 a8 38 66 8 0 82 43 b 3f 0 3 94 e7 79 6a 0 0 0 0 96 55 7 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319904437 us  
ping: Warning: time of day goes back (-274375018682545 s), taking countermeasures  
64 bytes from 192.168.56.102: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x10 but was 0xc0  
#16  c0 a8 38 66 8 0 82 43 b 3f 0 3 94 e7 79 6a 0 0 0 0 96 55 7 0 0 0 0 10 11 12 13  
#48  14 15 16 17 18 19 1a 1b  
--- 192.168.56.102 ping statistics ---  
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 3128ms  
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4  
$
```

File-Sharing and Remote Management

The scan should be examined for services associated with Windows file sharing and remote administration.

If present in the actual results, examples may include:

- SMB
- NetBIOS
- RDP
- RPC
- WinRM

Only services actually detected by the Nmap scan should be included in the final table.

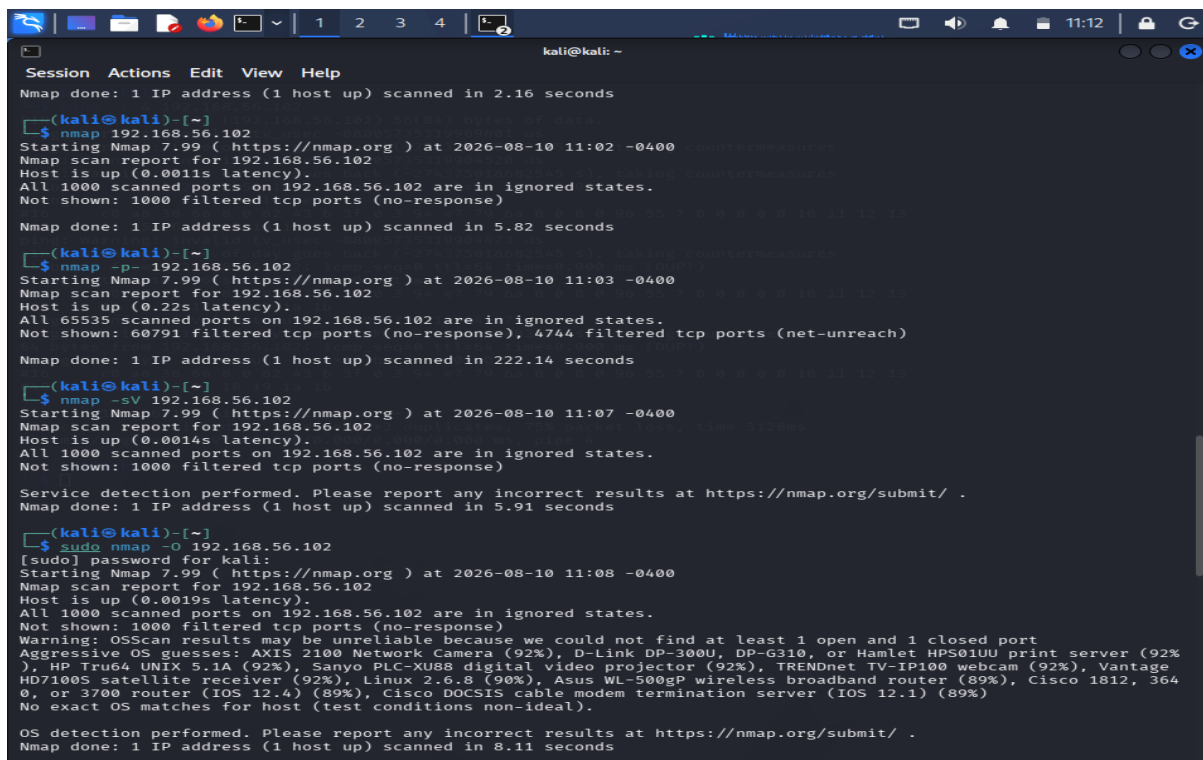
Least Privilege Evaluation

The principle of least privilege means that a workstation should expose only the network functions required for its intended role.

If a workstation has unnecessary file-sharing or remote-management services accessible from a broad network, its attack surface becomes larger. These services should therefore be restricted to trusted systems and authorized users.

Security Impact

Network enumeration gives an attacker useful information about the workstation's operating environment. Exposed remote-management or file-sharing services could potentially be abused if authentication, patching, access control, or firewall configuration is weak.



```
kali@kali: ~  
Session Actions Edit View Help  
Nmap done: 1 IP address (1 host up) scanned in 2.16 seconds  
  
[kali@kali]~$ nmap -p- 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:02 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0011s latency).  
All 1000 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Nmap done: 1 IP address (1 host up) scanned in 5.82 seconds  
  
[kali@kali]~$ nmap -sV 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:03 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.22s latency).  
All 65535 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 60791 filtered tcp ports (no-response), 4744 filtered tcp ports (net-unreach)  
Nmap done: 1 IP address (1 host up) scanned in 222.14 seconds  
  
[kali@kali]~$ nmap -sV 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:07 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0014s latency).  
All 1000 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 5.91 seconds  
  
[kali@kali]~$ sudo nmap -O 192.168.56.102  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:08 -0400  
Nmap scan report for 192.168.56.102  
Host is up (0.0019s latency).  
All 1000 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP True64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Vantage HD7100S satellite receiver (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)  
No exact OS matches for host (test conditions non-ideal).  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 8.11 seconds
```

This makes service minimization particularly important before the workstation enters a production network.

Recommended Hardening

1. Turn off unnecessary network services.
2. Configure Windows Firewall to restrict unwanted inbound connections.
3. Limit SMB access to authorized systems.
4. Restrict RDP to approved administrators.
5. Disable legacy protocols where they are not required.
6. Keep Windows and installed software updated.
7. Use strong account security.
8. Restrict remote management to designated administration systems.
9. Monitor remote-access events.
10. Use network segmentation where necessary.

Conclusion

The Windows workstation should expose only the services required for its intended purpose. Nmap enumeration provides a practical way to verify network exposure and identify services that need to be disabled or restricted before production deployment.

QUESTION 4

Enumeration of OWASP Juice Shop Server – 192.168.56.103

Aim

To determine the network exposure of the OWASP Juice Shop environment and identify the web service, supporting services, versions, and operating-system information before performing authorized penetration testing.

Commands Used

```
ping -c 4 192.168.56.103
```

```
nmap -sn 192.168.56.103
```

```
nmap 192.168.56.103
```

```
nmap -p- 192.168.56.103
```

```
nmap -sV 192.168.56.103
```

```
sudo nmap -O 192.168.56.103
```

```
nmap -sC 192.168.56.103
```

```
sudo nmap -sC -sV -O 192.168.56.103
```

Web Service Identification

If the actual scan identifies port 80:

```
nmap -p 80 --script http-title,http-headers 192.168.56.103
```

If the actual scan identifies port 443:

```
nmap -p 443 --script http-title,http-headers 192.168.56.103
```

Host Information

Target: 192.168.56.103

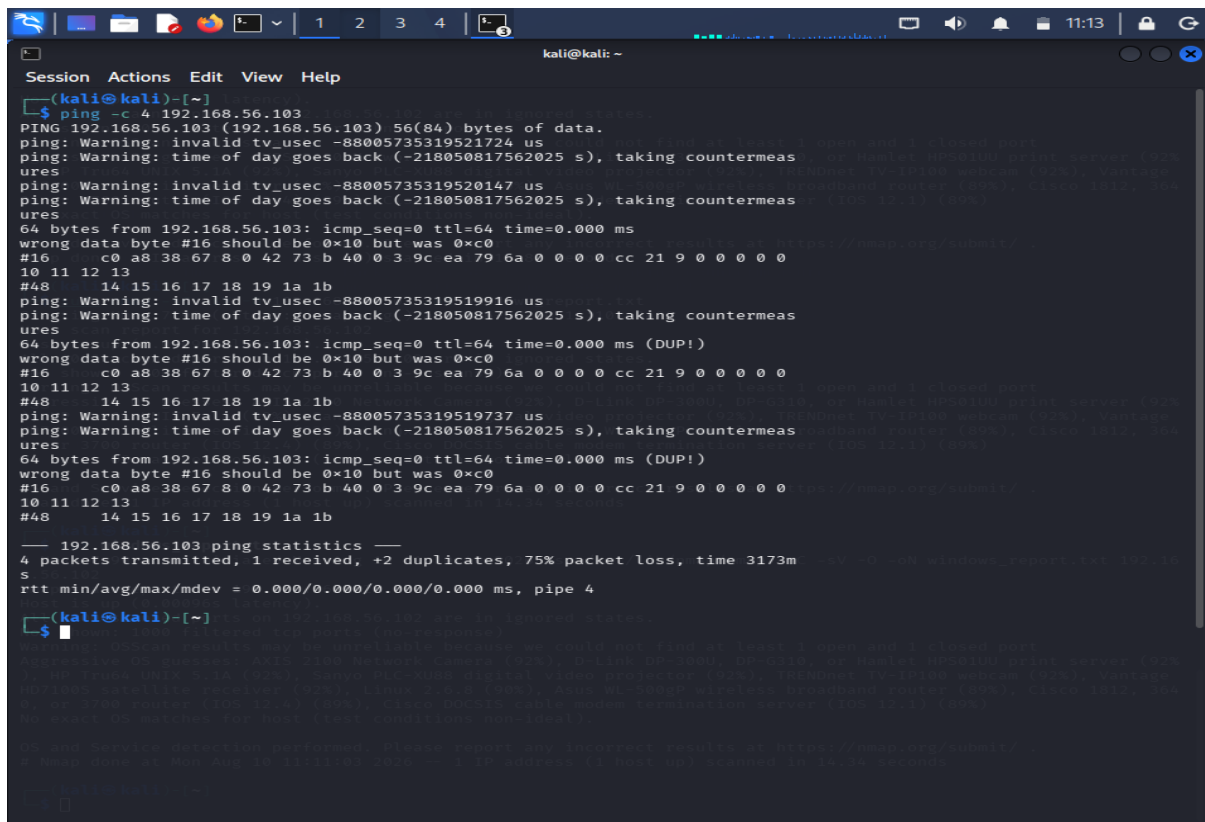
Service Results

Port	Service	Version	Role	Risk
[Actual]	[Actual]	[Actual]	Web application	[Level]
[Actual]	[Actual]	[Actual]	Supporting service	[Level]
[Actual]	[Actual]	[Actual]	Management/other	[Level]

Analysis

The web service is the primary network component required to make the Juice Shop application accessible. However, supporting services should not automatically be exposed to every network.

The Nmap results provide information about the application's hosting environment. Service names and versions can help security testers understand what technologies are present and prioritize areas for authorized testing.



```
kali@kali: ~  
Session Actions Edit View Help  
└─(kali@kali)-[~]  
└─$ ping -c 4 192.168.56.103  
PING 192.168.56.103 (192.168.56.103) 56(84) bytes of data.  
ping: Warning: invalid tv_usec -88005735319521724 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
ping: Warning: invalid tv_usec -88005735319520147 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms  
wrong data byte #16 should be 0x10 but was 0xc0  
#16 c0 a8 38 67 8 0 42 73 b 40 0 3 9c ea 79 6a 0 0 0 0 cc 21 9 0 0 0 0 0  
10 11 12 13  
#48 14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319519916 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x10 but was 0xc0  
#16 c0 a8 38 67 8 0 42 73 b 40 0 3 9c ea 79 6a 0 0 0 0 cc 21 9 0 0 0 0 0  
10 11 12 13  
#48 14 15 16 17 18 19 1a 1b  
ping: Warning: invalid tv_usec -88005735319519737 us  
ping: Warning: time of day goes back (-218050817562025 s), taking countermeas  
ures  
64 bytes from 192.168.56.103: icmp_seq=0 ttl=64 time=0.000 ms (DUP!)  
wrong data byte #16 should be 0x10 but was 0xc0  
#16 c0 a8 38 67 8 0 42 73 b 40 0 3 9c ea 79 6a 0 0 0 0 cc 21 9 0 0 0 0 0  
10 11 12 13  
#48 14 15 16 17 18 19 1a 1b  
— 192.168.56.103 ping statistics —  
4 packets transmitted, 1 received, +2 duplicates, 75% packet loss, time 3173m  
s  
rtt min/avg/max/mdev = 0.000/0.000/0.000/0.000 ms, pipe 4  
└─$
```

Security Concerns

Potential concerns include:

- Unnecessary open ports
- Exposed administration interfaces
- Outdated supporting software
- Excessive service-banner information
- Weak network access restrictions
- Insecure configurations

Security Recommendations

- Keep only application-required ports accessible.
- Restrict management interfaces.
- Patch the operating system and supporting software.
- Use HTTPS where appropriate.
- Prevent direct external access to internal databases.

- Apply firewall rules.
- Monitor web-server logs.
- Review exposed services regularly.

```

kali@kali: ~
Session Actions Edit View Help
Nmap done: 1 IP address (1 host up) scanned in 0.95 seconds

(kali@kali)-[~]
$ nmap -p 443 --script http-title,http-headers 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:43 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0015s latency).

PORT      STATE SERVICE
443/tcp    filtered https

Nmap done: 1 IP address (1 host up) scanned in 1.39 seconds

(kali@kali)-[~]
$ sudo nmap -sC -sV -O 192.168.56.103 -oN juiceshop_report.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 11:44 -0400
Nmap scan report for 192.168.56.103
Host is up (0.00053s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.18 seconds

(kali@kali)-[~]
$ cat juiceshop_report.txt
# Nmap 7.99 scan initiated Mon Aug 10 11:44:42 2026 as: /usr/lib/nmap/nmap -sC -sV -O -oN juiceshop_report.txt 192.168.56.103
Nmap scan report for 192.168.56.103
Host is up (0.00053s latency).
All 1000 scanned ports on 192.168.56.103 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Mon Aug 10 11:44:55 2026 -- 1 IP address (1 host up) scanned in 13.18 seconds

(kali@kali)-[~]
$

```

Conclusion

The enumeration phase provides a clear picture of the Juice Shop server's network exposure. This information can be used to prioritize authorized penetration-testing activities and reduce unnecessary exposure through proper network and server configuration.

QUESTION 5

Enumeration of Network Gateway – 192.168.56.1

Aim

To identify network and management services exposed by the laboratory gateway and evaluate the security risks associated with unnecessary administrative interfaces.

Commands Used

```
ping -c 4 192.168.56.1
```

```
nmap -sn 192.168.56.1
```

```
nmap 192.168.56.1
```

```
nmap -p- 192.168.56.1
```

```
nmap -sV 192.168.56.1
```

sudo nmap -O 192.168.56.1

nmap -sC 192.168.56.1

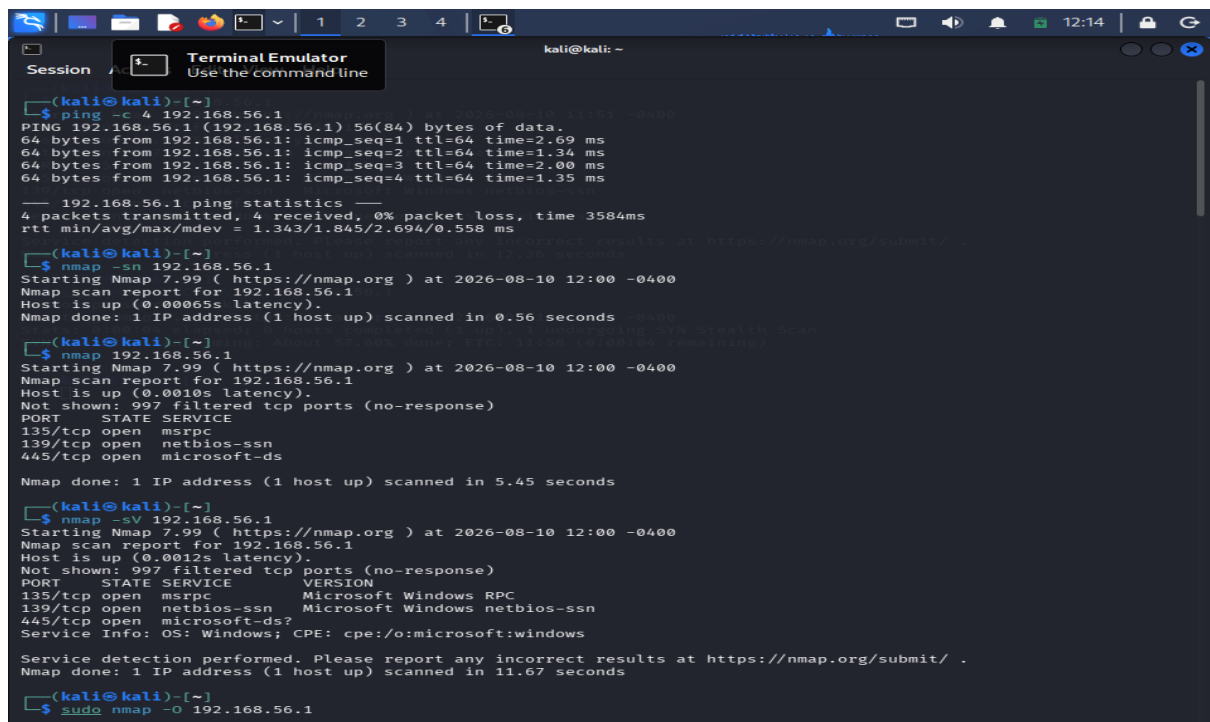
sudo nmap -sC -sV -O 192.168.56.1

For SNMP:

sudo nmap -sU -p 161 -sV 192.168.56.1

Host Information

Target: 192.168.56.1



```
(kali@kali)-[~]
└─$ ping -c 4 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data:
64 bytes from 192.168.56.1: icmp_seq=1 ttl=64 time=2.69 ms
64 bytes from 192.168.56.1: icmp_seq=2 ttl=64 time=1.34 ms
64 bytes from 192.168.56.1: icmp_seq=3 ttl=64 time=2.00 ms
64 bytes from 192.168.56.1: icmp_seq=4 ttl=64 time=1.35 ms
--- 192.168.56.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3584ms
rtt min/avg/max/mdev = 1.343/1.845/2.694/0.558 ms

(kali@kali)-[~]
└─$ nmap -sn 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:00 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00065s latency).
Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds

(kali@kali)-[~]
└─$ nmap 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:00 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0010s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
Nmap done: 1 IP address (1 host up) scanned in 5.45 seconds

(kali@kali)-[~]
└─$ nmap -sV 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:00 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0012s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.67 seconds

(kali@kali)-[~]
└─$ sudo nmap -O 192.168.56.1
```

Discovered Services

Port	Protocol	Service	Version	Function	Risk
[Actual]	TCP/UDP	[Actual]	[Actual]	[Function]	[Level]
[Actual]	TCP/UDP	[Actual]	[Actual]	[Function]	[Level]
[Actual]	TCP/UDP	[Actual]	[Actual]	[Function]	[Level]

Management Service Evaluation

SSH

SSH is designed for secure remote administration. However, exposing it to unrestricted networks can increase the number of systems that can attempt to access the gateway. It should therefore be limited to authorized administrators and trusted management hosts.

Telnet

Telnet does not provide adequate protection for modern administrative access. If found on the gateway, it should be removed where possible and replaced with a secure management protocol.

SNMP

SNMP supports monitoring and management. Improper configuration can potentially disclose useful information about the network device. Access should be restricted to approved monitoring systems.

HTTP

An HTTP management interface may expose administrative functionality without the protection expected from encrypted communication. It should be disabled where unnecessary.

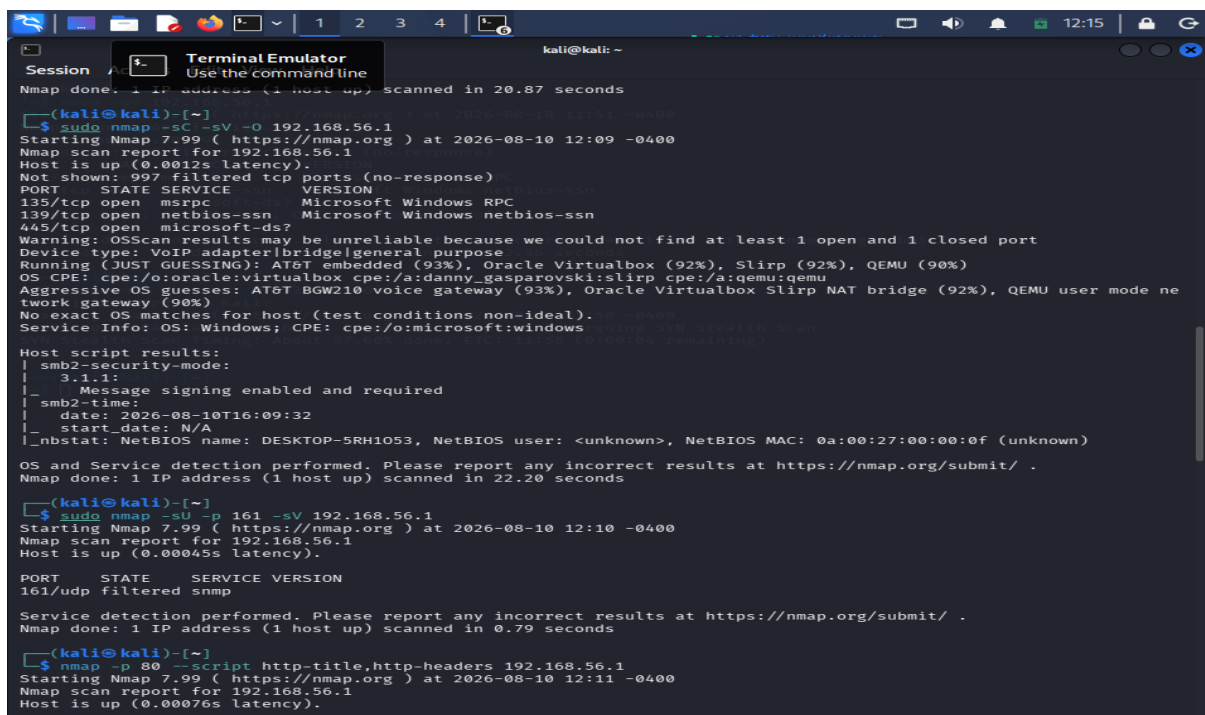
HTTPS

HTTPS provides encrypted web administration, but encryption alone does not make unrestricted management access safe. Access should still be limited to authorized management networks.

Risk Analysis

A gateway is a critical infrastructure component because it controls or facilitates network communication. An unnecessary management service can therefore represent a significant security concern.

Service enumeration helps administrators determine which interfaces are available and whether each one is justified by a legitimate operational requirement.



```
(kali@kali)-[~]
$ sudo nmap -sC -sV -o 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:09 -0400
Nmap scan report for 192.168.56.1
Host is up (0.0012s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
135/tcp    open  msrpc      Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb2-security-mode:
|   3.1.1:
|     Message signing enabled and required
|_ smb2-time:
|   date: 2026-08-10T16:09:32
|_ start_date: N/A
|_ nbstat: NetBIOS name: DESKTOP-5RH1053, NetBIOS user: <unknown>, NetBIOS MAC: 0a:00:27:00:00:0f (unknown)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.87 seconds

(kali@kali)-[~]
$ sudo nmap -sU -p 161 -sV 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:10 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00045s latency).
PORT      STATE SERVICE VERSION
161/udp    filtered snmp

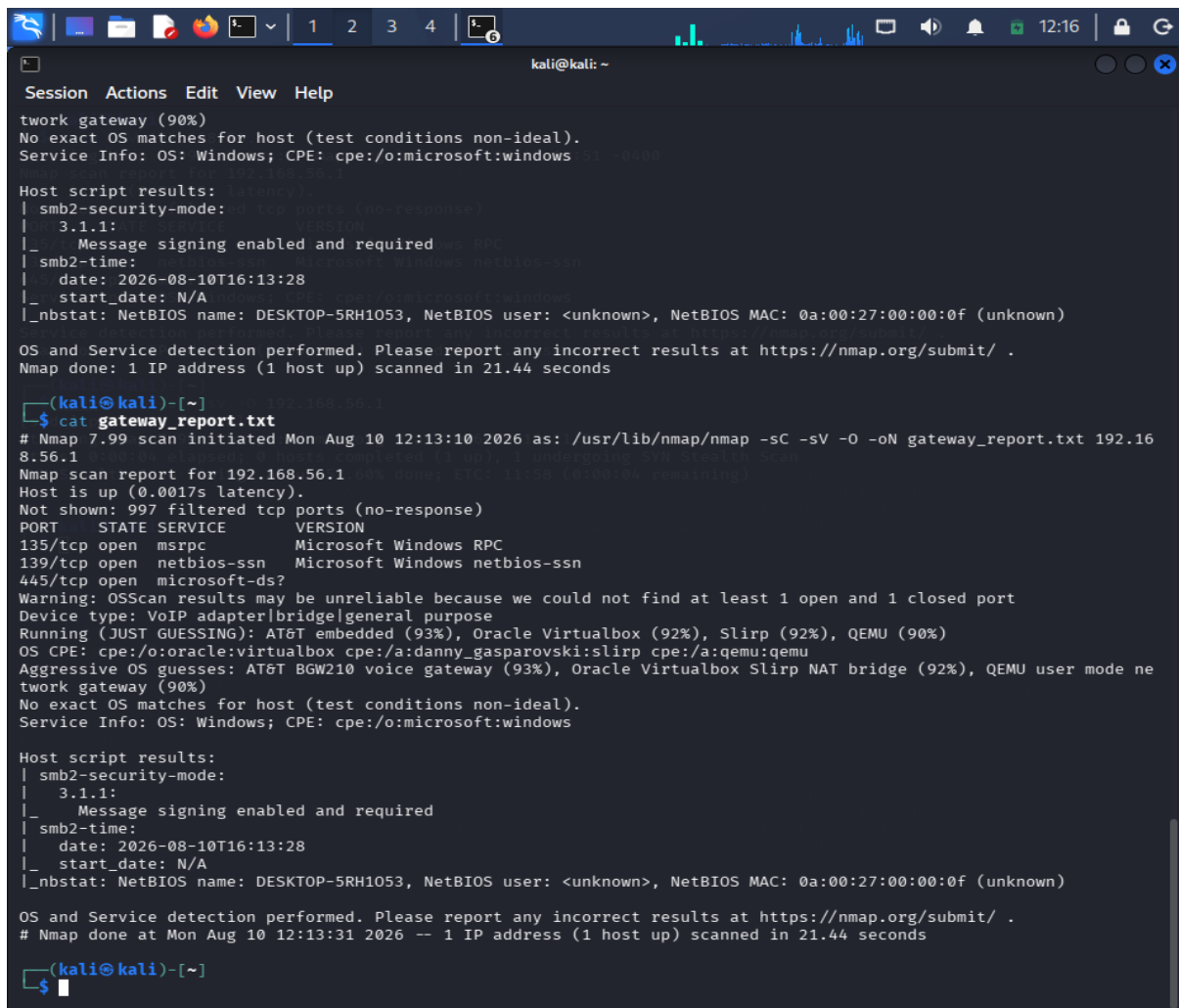
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.79 seconds

(kali@kali)-[~]
$ nmap -p 80 --script http-title,http-headers 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-10 12:11 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00076s latency).
```

Security Controls

1. Remove unused management services.

2. Disable Telnet.
3. Restrict SSH access.
4. Use strong administrator authentication.
5. Restrict SNMP access.
6. Prefer secure SNMP configurations.
7. Disable HTTP administration where possible.
8. Use properly configured HTTPS.
9. Keep gateway firmware updated.
10. Implement ACLs and firewall restrictions.
11. Monitor administrative activity.
12. Review gateway configuration periodically.



```
kali@kali: ~  
Session Actions Edit View Help  
work gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows 5.1.0.0000  
  
Host script results:  
| smb2-security-mode:  
| 3.1.1:  
|_ Message signing enabled and required  
| smb2-time:  
| date: 2026-08-10T16:13:28  
|_ start_date: N/A  
|_ nbstat: NetBIOS name: DESKTOP-5RH1053, NetBIOS user: <unknown>, NetBIOS MAC: 0a:00:27:00:00:0f (unknown)  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 21.44 seconds  
  
-(kali@kali)-[~]  
$ cat gateway_report.txt  
# Nmap 7.99 scan initiated Mon Aug 10 12:13:10 2026 as: /usr/lib/nmap/nmap -sC -sV -O -oN gateway_report.txt 192.168.56.1  
Nmap scan report for 192.168.56.1  
Host is up (0.0017s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
135/tcp   open  msrpc        Microsoft Windows RPC  
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn  
445/tcp   open  microsoft-ds?   
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)  
No exact OS matches for host (test conditions non-ideal).  
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows  
  
Host script results:  
| smb2-security-mode:  
| 3.1.1:  
|_ Message signing enabled and required  
| smb2-time:  
| date: 2026-08-10T16:13:28  
|_ start_date: N/A  
|_ nbstat: NetBIOS name: DESKTOP-5RH1053, NetBIOS user: <unknown>, NetBIOS MAC: 0a:00:27:00:00:0f (unknown)  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
# Nmap done at Mon Aug 10 12:13:31 2026 -- 1 IP address (1 host up) scanned in 21.44 seconds  
  
-(kali@kali)-[~]  
$
```

Conclusion

The gateway enumeration provides visibility into its exposed management interfaces and network services. Limiting administrative access, disabling insecure protocols, updating the gateway, and applying network-level restrictions can significantly reduce the possibility of unauthorized access.